INCH-POUND
MIL-DTL-24215/9D
24 July 2017
SUPERSEDING
MIL-DTL-24215/9C
w/AMENDMENT 1
15 August 2007

DETAIL SPECIFICATION SHEET

ATTENUATORS, VARIABLE (COAXIAL), CLASS 5 (SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the attenuators described herein shall consist of this specification and MIL-DTL-24215.

REQUIREMENTS

Dimensions and configuration: See figures 1, 2, 3, 4, 5, and 6.

Nominal impedance: 50 ohms.

Final finish: Flat black, in accordance with SAE-AMS-STD-595/37038.

Electrical requirements: See table I.

Part or identifying number (PIN): See table I.

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TABLE I. Electrical requirements.

PIN M24215/9-	Fig.	Freq. range (GHz)	Atten. range (dB) 3/	VSWR (max)	Insertion loss (max. at min. atten. setting) (dB)	Calibration accuracy (dB)	Frequency sensitivity (dB)	Min. input (watts)		Nominal operating temp. range (°C)	Weight (max)		
								av.	pk.		lbs	(g)	oz
001	1	2 - 8	0-15 at 2 GHz 0 -20 at 4-8 GHz	1.50:1	0.5	N/A	---	2	---	-45 - +100 (See notes 1 and 2)	.18	(82)	(2.9)
002	2	2 - 8	0-15 at 2 GHz 0-20 at 4-8 GHz	1.50:1	0.5	N/A	---	2	---	-45 - +100 (See notes 1 and 2)	.16	(73)	(2.5)
003	3	7 - 11	0 - 20	1.50:1	0.5	N/A	± 1.5	2	---	-45 - +100 (See notes 1 and 2)	.16	(73)	(2.5)
004	4	9.0 - 9.2	0 - 30	1.50:1	0.7	± 0.5	± 0.3	2	200	-18 - +70	1	(454)	(16)
005	5	4.0 - 18	0 - 15	1.50:1	1.0	N/A	± 2	5	3000	-30 - +105	.5	(227)	(8)
006	6	11 - 17	0 - 20	1.50:1	0.5	N/A	± 2	5	3000	-30 - +105	.16	(73)	(2.5)

NOTES:

1. Non-operating ambient temperature range: -60°C to +110°C.
2. Temperature coefficient: 1.0 dB maximum change over the temperature range -45°C to +55 °C at any attenuation setting.
3. Continuously variable.

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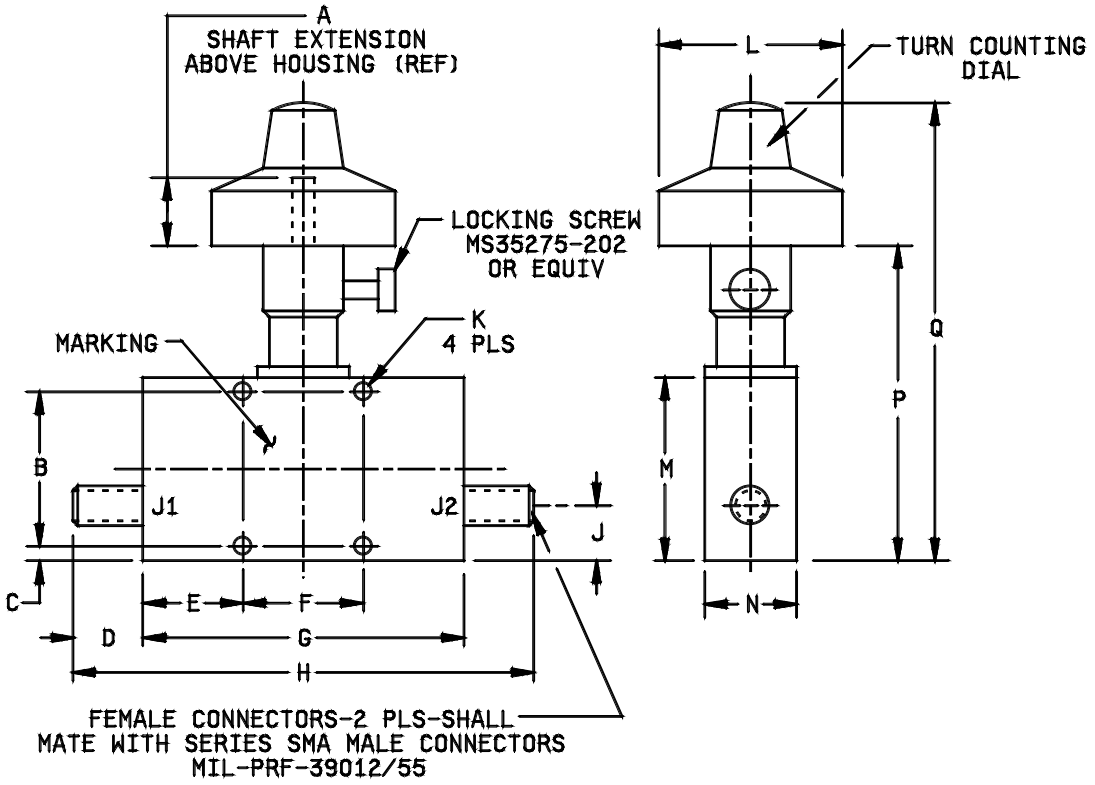


FIGURE 1. Dimensions and configuration for PIN M24215/9-001.

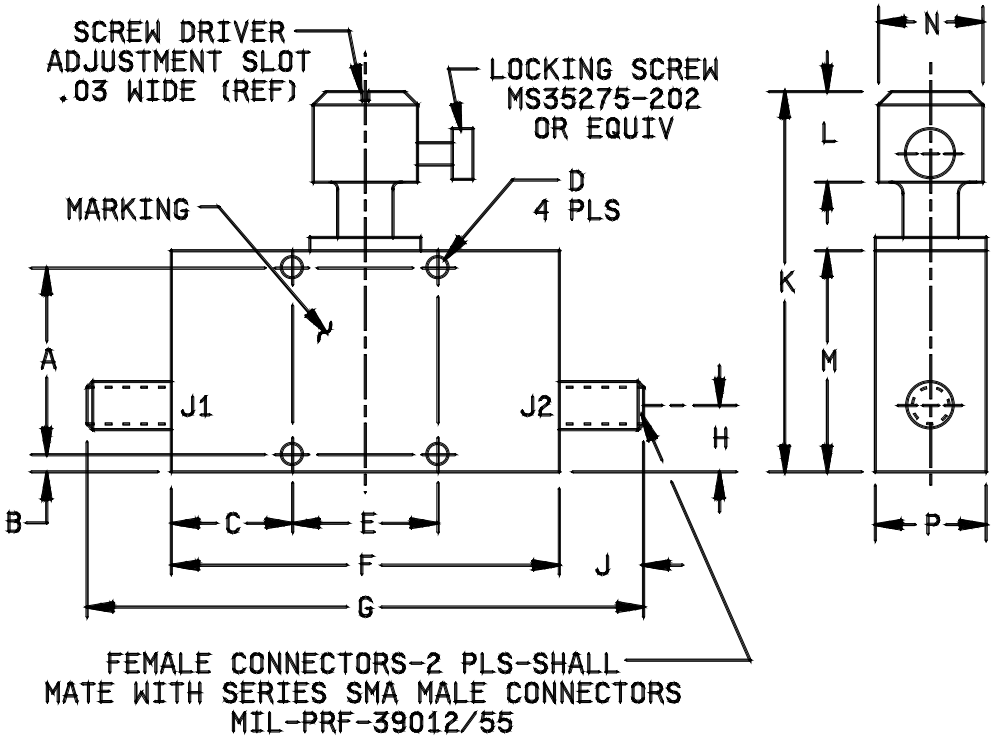
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Ltr	Inches		Millimeters	
	Max	Min	Max	Min
A	.58	.54	14.7	13.7
B	.849	.839	21.56	21.31
C	.11	.06	2.8	1.5
D	.40	.36	10.2	9.1
E	.58	.52	14.7	13.2
F	.661	.651	16.79	16.54
G	1.78	1.72	45.2	43.7
H	2.54	2.48	64.5	63.0
J	.33	.27	8.4	6.9
K	.107 dia	.101 dia	2.72 dia	2.57 dia
L	1.03 dia	.97 dia	26.2 dia	24.6 dia
M	1.03	.97	26.2	24.6
N	.53	.47	13.5	11.9
P	1.75	1.69	44.4	42.9
Q	2.53	2.47	64.3	62.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration for PIN M24215/9-001 - Continued.

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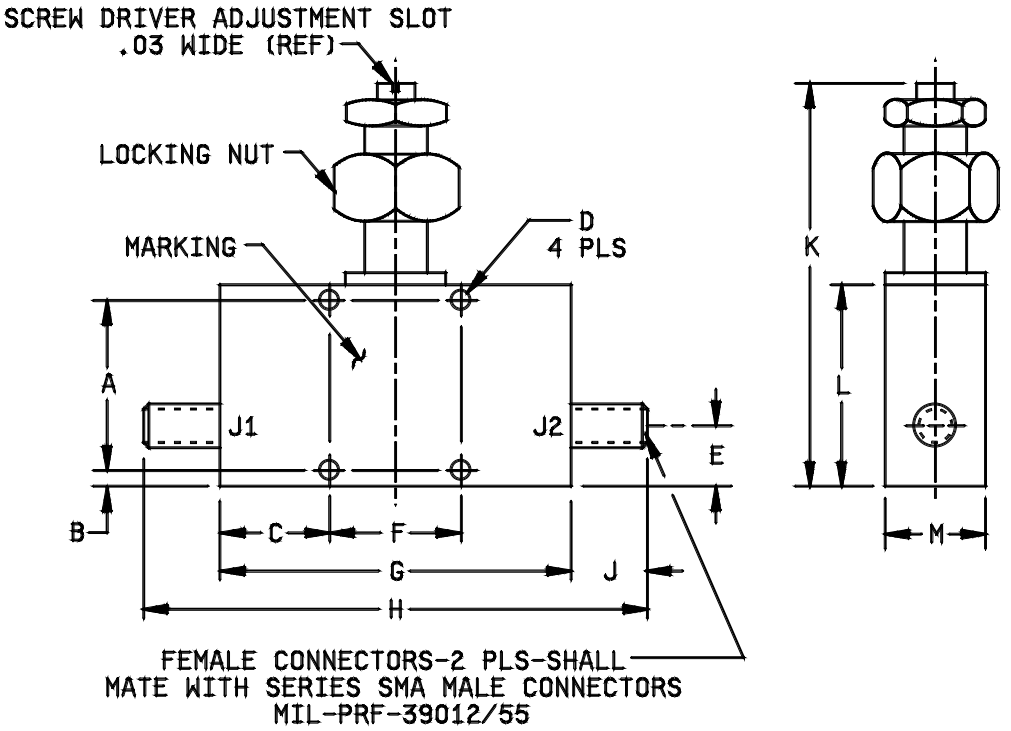


Ltr	Inches		Millimeters	
	Max	Min	Max	Min
A	.849	.839	21.56	21.31
B	.11	.06	2.8	1.5
C	.58	.52	14.7	13.2
D	.107 dia	.101 dia	2.72 dia	2.57 dia
E	.661	.651	16.79	16.54
F	1.78	1.72	45.2	43.7
G	2.54	2.48	64.5	63.0
H	.303	.297	7.70	7.54
J	.41	.35	10.4	8.9
K	1.75	1.69	44.4	42.9
L	.46	.36	11.7	9.1
M	1.03	.97	26.2	24.6
N	.52	.42	13.2	10.7
P	.53	.47	13.5	11.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Dimensions and configuration for PIN M24215/9-002.

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Ltr	Inches		Millimeters	
	Max	Min	Max	Min
A	.849	.839	21.56	21.31
B	.11	.06	2.8	1.5
C	.58	.52	14.7	13.2
D	.107 dia	.101 dia	2.72 dia	2.57 dia
E	.303	.297	7.70	7.54
F	.661	.651	16.79	16.54
G	1.78	1.72	45.2	43.7
H	2.54	2.48	64.5	63.0
J	.40	.36	10.2	9.1
K	2.03	1.97	51.6	50.0
L	1.03	.97	26.2	24.6
M	.53	.47	13.5	11.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Dimensions and configuration for PIN M24215/9-003.

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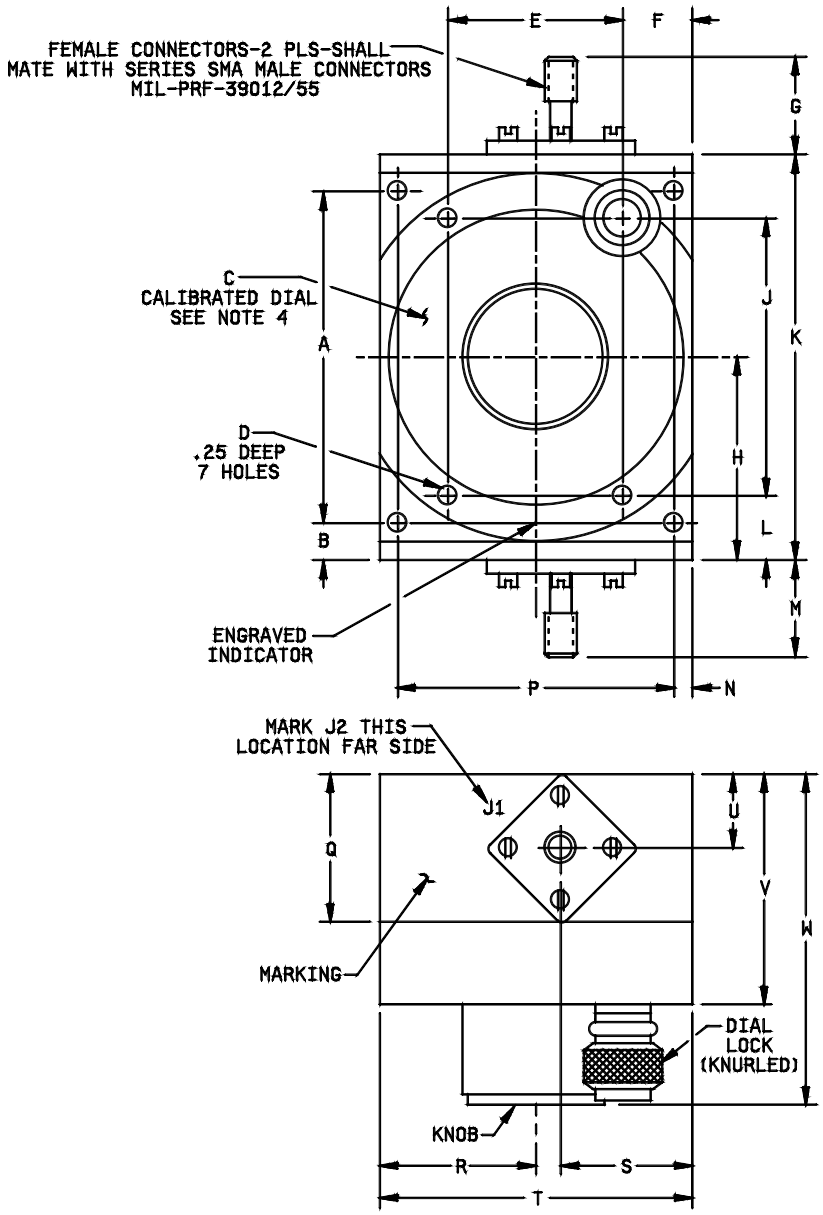


FIGURE 4. Dimensions and configuration for PIN M24215/9-004.

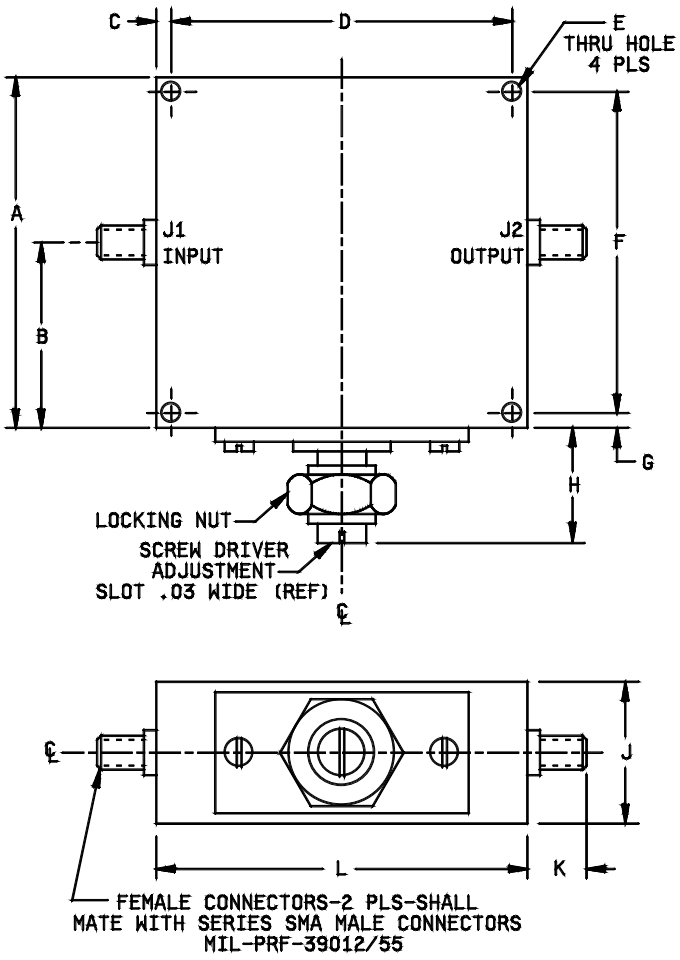
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	Inches		Millimeters	
	Max	Min	Max	Min
A	2.260	2.240	57.4	56.9
B	.27	.23	6.9	5.8
C	2.02 dia	1.98 dia	51.3 dia	50.3 dia
D	.27	.23	6.9	5.8
E	1.197	1.177	30.40	29.90
F	.49	.45	12.4	11.4
G	.68	.64	17.3	16.3
H	1.40	1.36	35.6	34.5
J	1.885	1.865	47.88	47.37
K	2.77	2.73	70.4	69.34
L	.46	.42	11.7	10.7
M	.68	.64	17.3	16.3
N	.14	.10	3.6	2.5
P	1.885	1.865	47.9	47.4
Q	1.03	.97	26.2	24.6
R	1.08	1.04	27.4	26.4
S	.90	.86	22.9	21.8
T	2.14	2.10	54.4	53.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.02 (.5 mm).
4. Dial shall be marked in one dB steps to the 10 dB level, and in two dB steps through the 30 dB level.

FIGURE 4. Dimensions and configuration for PIN M24215/9-004 - Continued.

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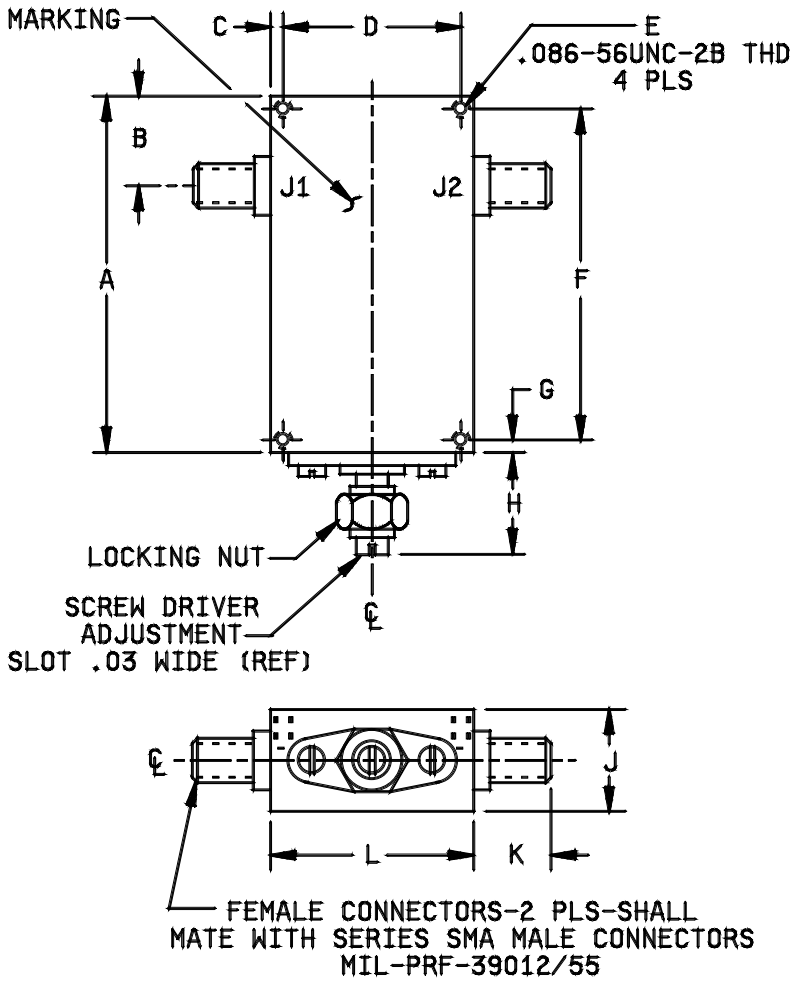


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

	Inches		Millimeters	
	Max	Min	Max	Min
A	2.28	2.22	57.9	56.4
B	1.22	1.16	31.0	29.5
C	.12	.06	3.0	1.5
D	2.192	2.182	55.7	55.4
E	.133 dia	.127 dia	3.38 dia	3.23 dia
F	2.067	2.057	52.5	52.2
G	.12	.07	3.0	1.8
H	.65	.59	16.5	15.0
J	.94	.88	23.9	22.4
K	.41	.35	10.4	8.9
L	2.41	2.35	61.2	59.7

FIGURE 5. Dimensions and configuration for PIN M24215/9-005

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	Inches		Millimeters	
	Max	Min	Max	Min
A	1.78	1.72	45.2	43.7
B	.47	.41	11.9	10.4
C	.09	.03	2.3	.8
D	.905	.845	22.99	21.46
E	---	.190 deep	---	4.83 deep
F	1.655	1.595	42.04	40.51
G	.09	.03	2.3	.8
H	.53	.47	13.5	11.9
J	.53	.47	13.5	11.9
K	.41	.35	10.4	8.9
L	1.03	.97	26.2	24.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.03 (.8 mm).

FIGURE 6. Dimensions and configuration for PIN M24215/9-006.

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Referenced documents: In addition to MIL-DTL-24215, this document references MIL-PRF-39012/55 and SAE-AMS-STD-595/37038.

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